

Project Design Phase Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between campus wellness/operational problems and technology solutions. For this project, it describes how raw data moves seamlessly from survey collection points to clean data models, and finally to high-fidelity, interactive dashboards designed for end-user intervention.

1. Technical Solution Breakdown

A. Data Ingestion Layer

- **Source A (Student Survey Data):** Captures multi-dimensional metrics including demographics, meal frequencies (skipping breakfast), stress levels, calorie awareness, and fast food preferences.
- **Source B (Optional Cafeteria Sales Data):** Captures transaction logs of menu selections, combos, and ingredient performance over time.
- **Storage:** Raw data files (CSV/Excel format) securely staged in an organizational directory or cloud storage bucket.

B. Data Preparation & Transformation Layer (ETL)

- **Tooling:** Tableau Prep Builder / Excel Power Query.
- **Processes:**
 - Data cleansing (handling missing values for meal choices, stripping outliers).
 - Data restructuring (pivoting survey matrices to individual behavioral profiles).
 - Feature mapping (correlating categorical variables like "Stress Level" with "Comfort Food Type").

C. Analytics & Analytics Engine Layer

- **Engine:** Tableau Hyper Extract Engine.
- **Process:** Compresses and optimizes data layers into ultra-fast query snapshots, enabling instant, cross-filtering performance when users toggle dashboard filters.

D. Presentation & Intervention Layer (Tableau Dashboards)

- **Dashboard 1: Student Self-Awareness Portal (For Rahul & Priya):** High-impact interactive workspace matching diet quality to perceived cognitive performance, stress-triggers, and energy trends.
- **Dashboard 2: Campus Menu Optimizer (For the Cafeteria Manager):** Strategic analytics view breaking down veggie/fruit consumption margins vs. fast food volumes to back new menu item launches.

Example - Solution Architecture Diagram:

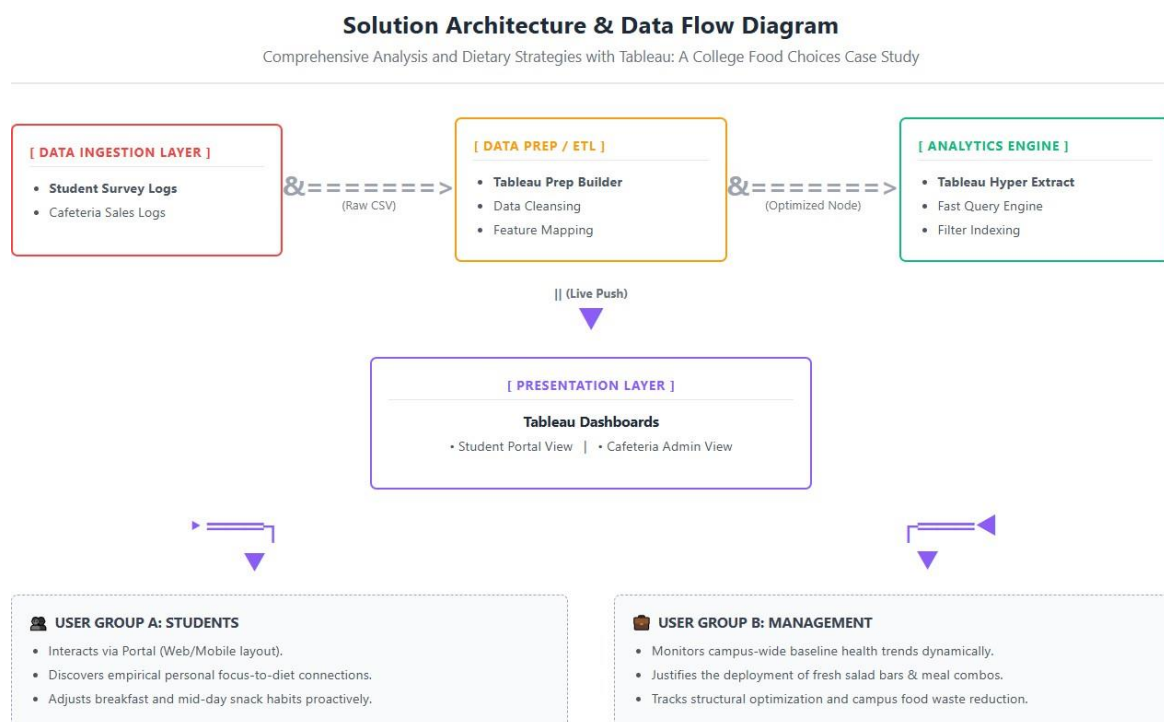


Figure 1: Comprehensive Analysis and Dietary Strategies with Tableau sample architecture

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>